Sprout Vale — Technical Sheet

Tech used

Single self-contained HTML file (about 30 KB), no external assets or libraries. HTML5 Canvas 2D with devicePixelRatio scaling (capped at 2x), gradient-shaded crops with glow, swaying plants, animated pond ripples, seasonal palettes, rain/storm/snow, butterflies, fireflies at night, a day-night lighting overlay, particles and floating numbers.

Web Audio API: every sound effect is synthesized, plus a procedural soundtrack whose scale and root change each season and drop an octave at night. Audio starts on first tap (iOS-safe) and suspends when the tab is hidden.

Pointer Events with drag-to-paint tiles, big 58px bottom-bar buttons, safe-area insets for notched phones, auto-save to localStorage every 5 seconds and each night, and a best-year record.

Game systems

8x10 tile farm with random rocks and a pond. Tile states: grass, rock, soil, watered, crop age, withered, placed objects.

8 crops across 4 seasons, single-harvest and regrowing types, 15% chance of a double harvest. Energy and clock costs per action, 2am pass-out penalty.

Overnight simulation: shipping, Order Board (double pay, 4-day deadline), eggs, growth, season change and withering, weather forecast (sun, rain, storm, snow), sprinklers, crows vs scarecrow coverage, storm damage.

Shop with 5 upgrades, placement and pick-up, medal tiers and save/continue.

Audit 1 — Balance (simulated play)

A bot played 30 full years at each of 3 skill levels (crops only; crops + sprinklers, scarecrows and energy; everything including chickens).

First pass: crop margins were too thin (median savings around 2,200g) and upgrades lowered a player's final gold, so buying anything felt like a trap. Then chickens at 55g eggs became dominant (median 10,700g).

Fixes: raised crop sale prices about 40%, cut sprinkler to 150g and stew to 220g, made chickens 300g with 40g eggs and a cap of 4, and tied the score to final savings.

Final results: skill 0 min 3476 med 5797 max 6806 crow 3.2 orders 2.3 broke 0; skill 1 min 2926 med 4589 max 6993 crow 2.7 orders 2.2 broke 0; skill 2 min 3485 med 6805 max 10115 crow 2.9 orders 2.2 broke 0. No run went broke, and the medal thresholds (3,500 / 6,000 / 9,000) spread across the skill levels.

Audit 2 — Bugs

Ran a headless smoke test of the real file: tilling, planting, watering, refilling, every menu, placing a scarecrow, sleeping, harvesting, the 2am pass-out, save and reload, a full year to the end screen and back to the title. No errors.

Fixed: when a season changed, the selected seed stayed on last season's crop, so every tap showed an error. The seed selection now switches to the new season's first crop.

Fixed: picking up a placed sprinkler or scarecrow with the Hoe on a grass tile tilled it instead; pick-up is now checked first.

Audit 3 — Player experience

Added a low-energy hint that reminds you harvesting is free, a midnight warning before you pass out, an "order ready" toast when your bag holds enough, and an always-visible order banner.

The seed list warns when a crop won't ripen before the season ends, and the bedtime screen shows tomorrow's forecast so you know whether to water.

The night report itemises every sale, the order payout, eggs, crow and storm losses, and explains the new season.